THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 10 **(b)** 4 **(c)** 1 **(d)** 7

2. The pH of gastric juice is about: 1

(a) 7.4 **(b)** 1.4 **(c)** 5.5 **(d)** 10

3. Milk of magnesia, an antacid, is: 1

(a) magnesium sulphate **(b)** magnesium hydroxide
(c) magnesium chloride **(d)** magnesium carbonate

4. A solution reacts with crushed egg-shells to give a gas that turns lime water milky. The solution contains: 1

(a) NaCl **(b)** Na_2SO_4 **(c)** HCl **(d)** NaOH

5. Tooth decay starts when the pH of the mouth falls below: 1

(a) 7 **(b)** 9 **(c)** 5.5 **(d)** 3

6. Phenolphthalein turns pink in: 1

(a) neutral solution **(b)** acidic solution **(c)** any solution **(d)** basic solution

7. 10 mL of NaOH solution is completely neutralised by 8 mL of HCl. If 20 mL of the same NaOH is taken, the HCl needed is: 1

(a) 8 mL **(b)** 4 mL **(c)** 16 mL **(d)** 12 mL

8. The acid present in tamarind is: 1

(a) lactic acid **(b)** tartaric acid **(c)** citric acid **(d)** oxalic acid

9. Methyl orange in acidic solution is: 1

(a) colourless **(b)** blue **(c)** yellow **(d)** red

10. Which type of medicine is used for treating indigestion? 1

(a) antibiotic **(b)** analgesic **(c)** antacid **(d)** antiseptic

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does dry HCl gas not change the colour of dry litmus paper? 2

12. Why does an aqueous solution of an acid conduct electricity? 2

13. Write the chemical name and formula of bleaching powder. 2

14. While diluting an acid, why is it recommended that the acid be added to water and not water to acid? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the general reaction of an acid with a metal. Give the reaction of dilute HCl with zinc and the test for the gas. 3

16. What is the chlor-alkali process? Name its three products and write the equation. 3

17. Write what happens when HCl dissolves in water. What is hydronium ion? 3

OR

What happens when a metal carbonate reacts with an acid? Write the reaction for Na_2CO_3 with HCl and the test for the gas. 3

18. How is bleaching powder prepared? State two uses. 3

OR

Solution A has pH 2 and solution B has pH 5. Which is more acidic? By how many times does the H^+ concentration differ? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the reaction of dilute HCl with zinc granules. State the observations, the test for the gas, the equation and the general reaction. 5

OR

Describe the chlor-alkali process. Write the equation, name the electrodes where gases are formed and give one use of each product. 5

20. Describe the activity to show that acids and glucose solutions behave differently in conducting electricity. State the conclusion. 5

OR

How are bleaching powder and baking soda prepared? Give two uses of each. 5

END OF QUESTION PAPER — Total: 40 Marks

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